

MapReduce in Hadoop using Python on Ubuntu - Complete Practical Guide

This guide explains step-by-step how to run simple MapReduce programs in Ubuntu using Python. It includes: 1. Character Counting 2. Word Counting 3. All Ubuntu terminal commands 4. File creation 5. Execution steps 6. Expected outputs

1. Install Python

Open Ubuntu terminal and run:

```
sudo apt update
sudo apt install python3 -y
```

Check Python version:

```
python3 --version
```

2. Create Working Folder

Create a folder for the practical:

```
mkdir mapreduce_practical
```

Go inside the folder:

```
cd mapreduce_practical
```

3. Create Input File

Create input text file:

```
nano input.txt
```

Example content for character counting:

```
HELLO
```

OR for word count:

```
hello world hello
```

Save file:

CTRL + O

Press ENTER

CTRL + X

4. Character Count - Mapper Code

Create mapper file:

```
nano mapper.py
```

Paste this code:

```
import sys

for line in sys.stdin:
    line = line.strip()
    for ch in line:
```

```
        if ch != " ":
            print(ch + "\t1")
```

Save and exit:
CTRL + O
ENTER
CTRL + X

5. Character Count - Reducer Code

Create reducer file:

nano reducer.py

Paste this code:

```
import sys

current_char = ""
current_count = 0

for line in sys.stdin:
    line = line.strip()

    ch, count = line.split("\t")

    count = int(count)

    if ch == current_char:
        current_count = current_count + count
    else:
        if current_char:
            print(current_char + "\t" + str(current_count))

        current_char = ch
        current_count = count

print(current_char + "\t" + str(current_count))

Save and exit:
CTRL + O
ENTER
CTRL + X
```

6. Run Character Count Program

Run this command:

```
python3 mapper.py < input.txt | sort | python3 reducer.py
```

Expected output:

```
E    1
H    1
L    2
O    1
```

7. Word Count - Mapper Code

Replace mapper.py content with:

```
import sys

for line in sys.stdin:
    line = line.strip()
```

```
words = line.split()

for word in words:
    print(word + "\t1")

Save and exit.
```

8. Word Count - Reducer Code

Replace reducer.py content with:

```
import sys

current_word = ""
current_count = 0

for line in sys.stdin:
    line = line.strip()

    word, count = line.split("\t")

    count = int(count)

    if word == current_word:
        current_count = current_count + count
    else:
        if current_word:
            print(current_word + "\t" + str(current_count))

        current_word = word
        current_count = count

print(current_word + "\t" + str(current_count))

Save and exit.
```

9. Run Word Count Program

Run this command:

```
python3 mapper.py < input.txt | sort | python3 reducer.py
```

Expected output:

```
hello    2
world    1
```

10. Important Viva Questions

Q1. What is Mapper?

Mapper converts input data into key-value pairs.

Q2. What is Reducer?

Reducer combines values having same keys.

Q3. What does sort do?

It groups same keys together.

Q4. What is Hadoop?

Hadoop is a distributed framework for big data processing.

Q5. What is MapReduce?

MapReduce is a programming model used for distributed processing.

11. One-Line Memory Trick

Mapper = Break and Emit
Reducer = Combine and Count